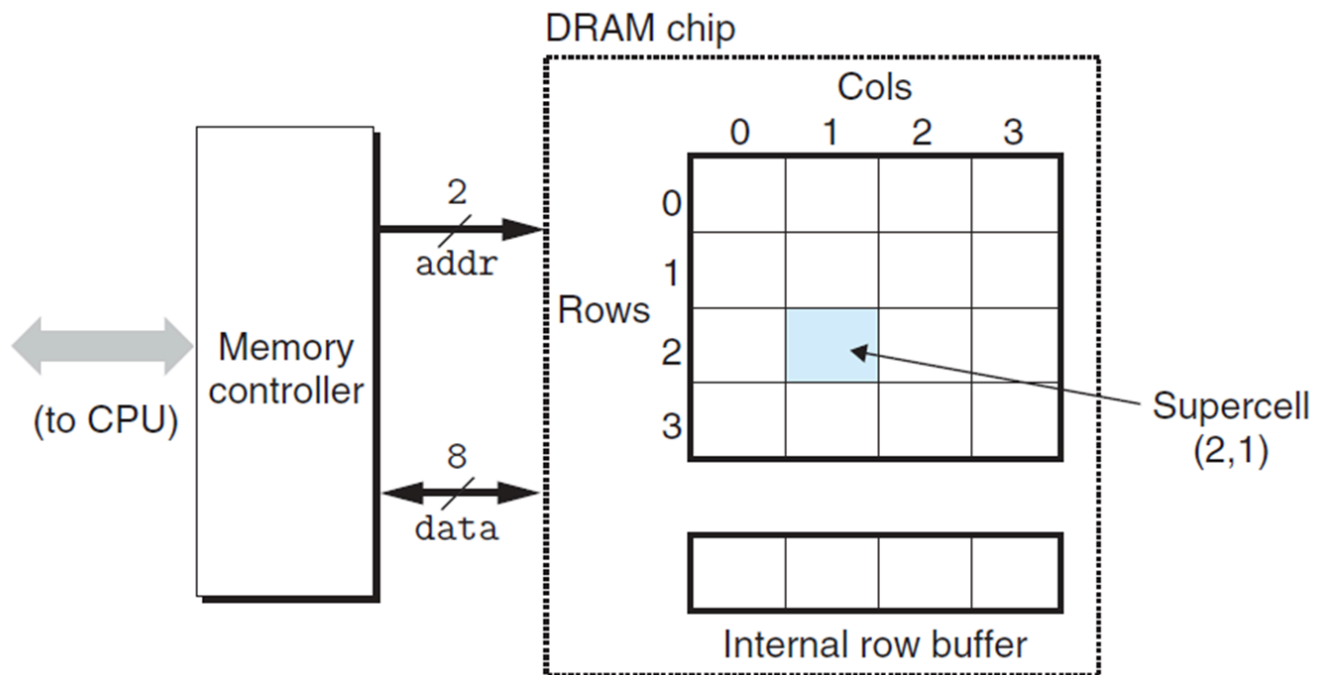


# Lecture 6 Memory Hierarchy

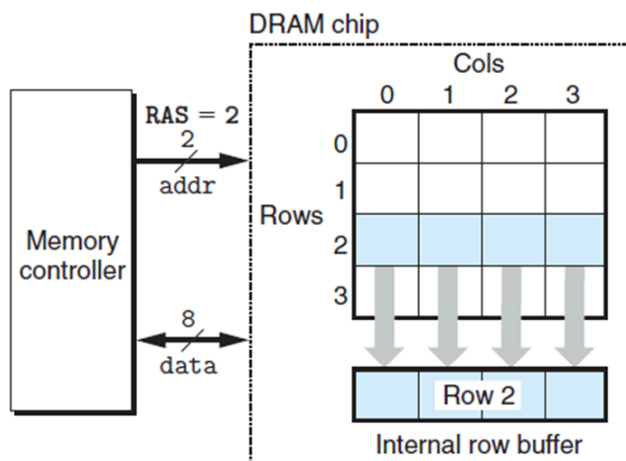
## Conventional DRAMs



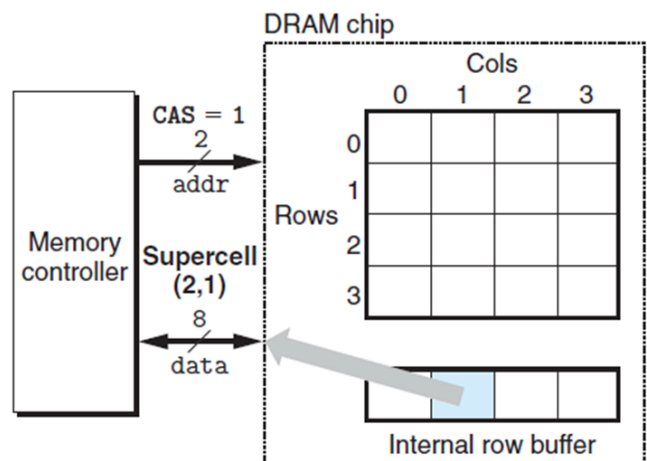
A **cell** is the smallest storage of 1 bit. The **supercell** is the addressable smallest unit, usually 8 bits or 1 byte as a supercell.

The **memory controller** communicates with CPU and DRAMs, translating addresses **between the virtual address and actual position of DRAM chip**.

The internal **row buffer** stores data of one complete row after accepting RAS. If **row hit** (access the same row next time), directly access row buffer to improve efficiency. If **row miss**, close current row and renew corresponding row.

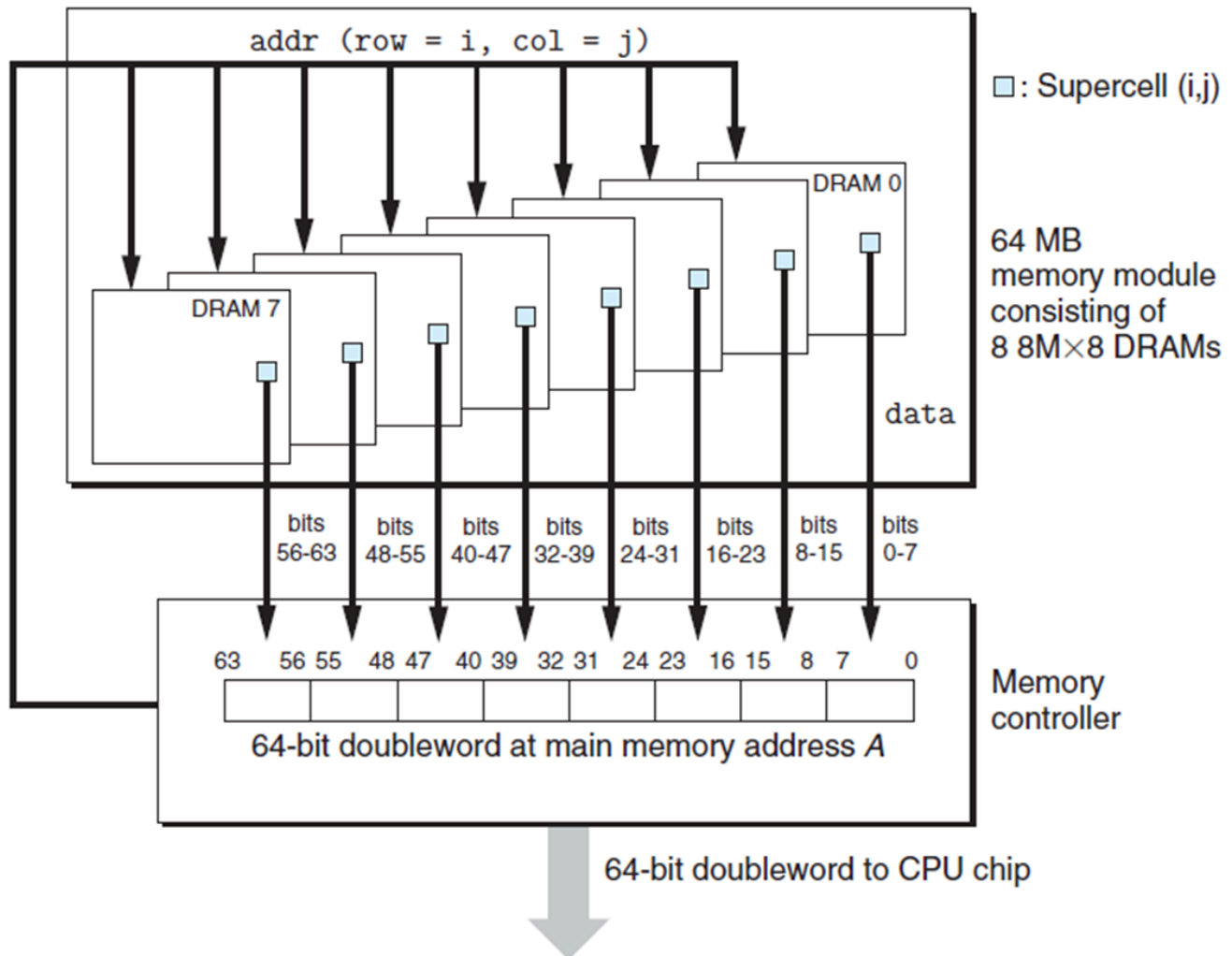


(a) Select row 2 (RAS request).



(b) Select column 1 (CAS request).

# Memory Modules



Obtain 1 byte from each DRAM with the same row and column to get the 64-bit virtual address or data.

## Locality

Principle of locality:

Programs tend to use **data and instructions** with **addresses near or equal to those they have used recently**.

- Temporal locality: recently referenced items are likely to be referenced again **in the near future**.
- Spatial locality: items with **nearby addresses** tend to be referenced close together in time.

## Locality Example

```
sum = 0;
for (i = 0; i < n; i++)
    sum += a[i];
return sum;
```

- Data reference:
  - Spatial locality: reference array elements in succession
  - Temporal locality: reference variable `sum` each iteration
- Instruction reference:
  - Spatial locality: Reference instructions **in sequence**.
  - Temporal locality: cycle through loop **repeatedly**.

## Classwork

Problem 6.8